



**EASTERN MEDITERRANEAN UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING**

**MENG/MECT 400 SUMMER PRACTICE
(FALL 2020-2021)**

Student's

**Name, Surname: TARIK
TARIQ**

Number:23704019

CATEGORIES		SO	WEIGHT /100	MARK
Record and Logbook Keeping		-	-	
Company's Supervisor Evaluation		4	50	
Report		3, 7		
Ch1	Introduction	-	10	
Ch2	Product Information	-	10	
Ch3	Performed Work	-	10	
Ch4	Personal Development and Awareness about Contemporary Issues	-	10	
Appendix	Standards		10	
TOTAL			100	

Student Outcomes		
1		an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2		an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3	X	an ability to communicate effectively with a range of audiences
4	X	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5		an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6		an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7	X	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

MENG/MECT400 SUMMER PRACTICE REPORT

Submitted to the Summer Practice Committee in Partial Fulfilment of the
Requirements for the Summer Practice Course MENG/MECT400

Fall 2020-2021

Name Surname, Student No:

TARIK TARIK,23704019

Eastern Mediterranean University

Faculty of Engineering

Department of Mechanical Engineering

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ACKNOWLEDGEMENT

wish to express my profound gratitude to Bamburi Cement Limited, Mombasa, for providing me with the invaluable opportunity to undertake my industrial attachment within the Mechanical Department from July to September 2025. The experience was instrumental in bridging the gap between theoretical knowledge and practical industrial application.

My sincere appreciation goes to the company's Health and Safety Team for the comprehensive induction and safety training, which instilled in me a deep respect for workplace safety protocols that guided all my activities. I am particularly indebted to my supervisor and the entire Mechanical Maintenance team for their exceptional mentorship and guidance. Their willingness to involve me in hands-on tasks from assisting in the disassembly of high-volume screw compressors and gearboxes to observing critical alignment procedures with laser systems—provided me with an unparalleled learning experience.

I extend special thanks to the Reliability Engineer for the insightful tutorial on Mean Time Between Failures (MTBF) and vibration analysis, which opened my eyes to the world of predictive maintenance. The knowledge gained from observing the use of thermocameras on the rotary kiln and spectrograms for vibration analysis has given me a significant understanding of modern, data-driven maintenance strategies.

Furthermore, I am thankful to all the technicians and fitters I shadowed across various sections, including the Crusher, Rotary Kiln, and Cement Mill. Their patience in explaining complex processes, from the journey of clinker production to the meticulous inspection of bucket elevators and conveyor systems, was the cornerstone of my learning.

This attachment has immensely enriched my technical competencies and professional outlook. The exposure to Bamburi Cement's commitment to operational excellence, environmental stewardship, and safety has set a benchmark for my future career. The skills and memories I am taking with me are invaluable, and for that, I am eternally thankful.

CHAPTER 1 INTRODUCTION

1.1 Background

Industrial attachment is a crucial component of engineering education, providing students with the opportunity to apply theoretical knowledge in practical settings. It bridges the gap between classroom learning and real-world industry practices, helping students to acquire technical skills, workplace discipline, and problem-solving abilities. I undertook my industrial attachment at **Bamburi Cement PLC** at the Mombasa plant, one of the most established cement manufacturing facilities in East Africa. This attachment exposed me to cement production processes, equipment operation, and maintenance procedures, allowing me to gain valuable professional experience in a large-scale industrial environment.

1.2 Company Information

Bamburi Cement PLC was established in 1951 and began production in 1954, growing to become one of the largest cement manufacturers in Kenya. The company operates two main facilities: an integrated plant in Mombasa and a grinding plant in Athi River. These plants serve both domestic and regional markets with a wide range of cement and concrete products.

In **December 2024**, Bamburi Cement officially became a member of the **Amsons Group**, a Tanzanian conglomerate with growing influence across East Africa. Amsons acquired a 58.6% majority stake previously held by Holcim, following shareholder approval. This transition marked a new phase for Bamburi, as Amsons has committed to investing in modernization, capacity expansion, and sustainable operations. The company remains a key player in Kenya's construction sector and continues to uphold high standards of quality, innovation, and environmental stewardship. The rehabilitation of former quarries into nature reserves such as **Haller Park** is one of its notable contributions to environmental sustainability.

1.3 Structure of the Report

This report is organized into four chapters. Chapter One introduces the company and outlines the purpose of the attachment. Chapter Two provides detailed information about the company's products. Chapter Three discusses the work performed during the attachment,

including daily activities and assigned tasks. Chapter Four highlights personal development and awareness of contemporary issues gained throughout the attachment period. The report concludes with references and appendices containing relevant supplementary information.

CHAPTER 2 PRODUCT INFORMATION

2.1 PRODUCTS OF THE COMPANY

1. CEMENT PRODUCTS

Bamburi Cement PLC produces a wide range of high-quality cement brands designed to meet the diverse needs of the construction industry. At the Mombasa plant, the major cement products include:

- **Tembo cement**

Tembo is Bamburi's flagship Portland Pozzolana Cement (PPC). It is blended with pozzolanic materials to enhance durability, making it highly resistant to aggressive environments such as coastal conditions. It is widely used in residential and commercial construction, plastering, and general masonry works because of its excellent workability and long-term strength.

- **Fundi cement**

Fundi is a masonry cement specifically designed for non-structural applications. It is ideal for plastering, rendering, mortar works, and block making. Its smooth finish, ease of application, and cost-effectiveness make it popular in housing and finishing projects.

- **PowerPlus cement**

PowerPlus is a high-strength Ordinary Portland Cement (OPC) suitable for demanding structural applications. It is widely used in infrastructure projects such as bridges, high-rise buildings, precast concrete, and heavy-duty foundations. It offers rapid early strength development and superior performance under high load conditions.

- **Nguvu Cement**

Nguvu is a multipurpose cement suitable for a wide range of general construction needs. It is commonly used in foundations, slabs, pavements, and other structural works where strength and durability are required. It provides an excellent balance between performance and affordability.



Figure 1 Cement Products produced by Bamburi cement

2. PRODUCTION PROCESS

The cement production process at the Mombasa plant follows four main stages:

- **Quarrying and Raw Material Preparation**

Coral, shell, bauxite and iron ore raw materials are extracted from company-owned quarries. The materials are crushed, blended, and homogenized to achieve the required chemical composition with coral consisting of about 80% of the composition.

- **Clinker Production**

The raw mix is heated in rotary kilns at temperatures around 1,400°C. This process produces clinker, the key intermediate material in cement production.

- **Cement Grinding**

Clinker is mixed with gypsum and, depending on the brand, pozzolana or other additives. The mixture is ground in vertical roller mills or ball mills to achieve the required fineness. The type and proportion of additives determine the characteristics of different cement brands (e.g., pozzolana for Tembo, masonry additives for Fundi).

- **Packaging and Distribution**

Finished cement is packed in 50 kg bags under brand names Tembo, Fundi, PowerPlus, and Nguvu. For bulk orders, cement is loaded directly into tankers for delivery to large projects.

3. Employed Technologies

The Mombasa plant operates with **modern rotary kilns, vertical roller mills, ball mills and automated packaging lines**. Process control is managed through **computerized systems** that ensure product consistency and efficiency. Environmental technologies such as **electrostatic precipitators and bag filters** are employed to reduce dust emissions, while alternative fuels and raw materials are increasingly being integrated to promote sustainability.

4. Departmental Interactions

Smooth operations at Bamburi Cement rely on close collaboration between different departments:

- The **Quarry Department** supplies Coral and other raw materials.
- The **Production Department** oversees clinkerization and cement grinding.
- The **Quality Control Laboratory** conducts continuous testing to ensure each cement brand meets both local and international standards.
- The **Engineering and Maintenance Department** ensures plant machinery operates reliably with preventive and corrective maintenance.
- The **Packaging and Logistics Department** manages bagging, storage, and distribution of Tembo, Fundi, PowerPlus, and Nguvu cement to customers across Kenya and the region.

CHAPTER 3 PERFORMED WORK

3.1 Daily Activities and Performed Work at the Workplace

During my summer attachment at Bamburi Cement PLC, Mombasa Plant, I was placed under the Mechanical Department, which is divided into the Methods and Execution sections. My daily activities involved observing, assisting, and actively participating in inspection, maintenance, and troubleshooting of various machines and systems. The experience provided me with both individual learning opportunities and teamwork engagements alongside engineers, inspectors, and technicians.

In the **first week**, I focused on orientation and administrative formalities. I reported to the company, received my personal protective equipment (PPE), and attended induction training on workplace safety, company policies, and operational procedures. I was introduced to my supervisor, team members, and the clinker manufacturing process. This gave me an overall understanding of the facility layout and reporting structure.

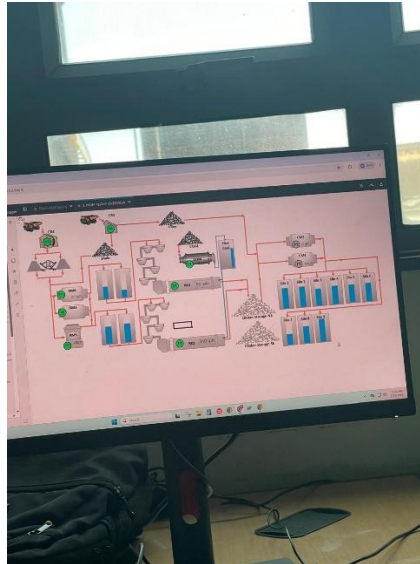


Figure 2 Entire manufacturing process of cement

From the **second week** onwards, I began participating in practical field activities. I observed the inspection of the coral weigh feeder chute, assisted in tightening bolts of a compressor delivery line that had a leak, and helped disassemble a high-volume screw compressor. I also learned how conveyor belt alignment is carried out and observed general inspections of raw

mill sections. Later in the week, I was introduced to the crusher section and the pre-blending hall, where I learned about the stacking and reclaiming processes of coral. I also participated in the disassembly of a low-speed gearbox, gaining exposure to its shafts and gear sets. These activities highlighted the importance of preventive maintenance in supporting continuous clinker production.

The **third week** exposed me to other critical sections such as the packing plant, where I learned about the operation of the roto packer system and how spillage or misalignment affects weighing accuracy. I visited the machine shop and observed machining of shafts and key seats. I also participated in vibration data collection using triaxial accelerometers on raw mill bearings and observed how spectrograms were analyzed to diagnose potential faults. In the rotary kiln section, I witnessed disassembly of a crusher motor, a fluid coupling, and a flywheel during bearing replacement. I later attended a working-at-heights safety training and observed the reassembly of the crusher motor using filler gauges and laser alignment tools. Towards the end of the week, I developed an understanding of the induced draft fan system, gas conditioning tower, and how evaporative cooling reduces gas temperature.



Figure 3 ROTO PACKER used for packaging cement



Figure 4 GCT (gas conditioning tower) where evaporative cooling of hot gases happens

In the **fourth week**, I shadowed technicians during mechanical inspections, assisting in tightening bolts on chain elevators and observing the maintenance of airlocks used to maintain pressure seals. I learned how clearance tolerances were achieved and observed troubleshooting in routine operations. I was also introduced to measuring slip in turbo couplings using a stroboscope, an essential practice to ensure slip remains below 4%. Later, I attended training on bearing fundamentals, where I learned about various bearing types, loading conditions, and nomenclature. This was followed by another session on bearing failures, where I understood the causes of failure, the role of lubrication, and the importance of proper film thickness for reliable operation.



Figure 5 turbo coupling

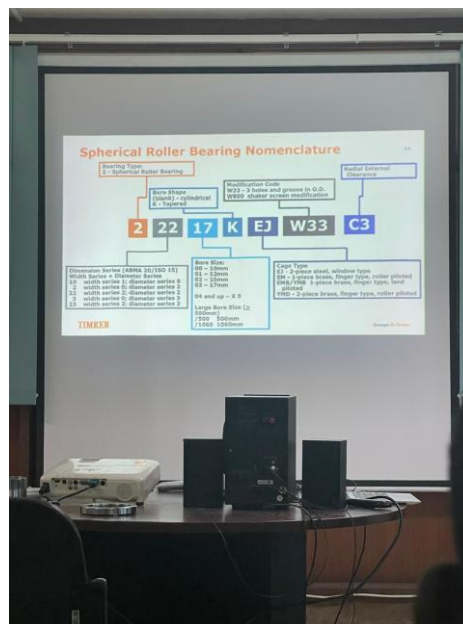


Figure 6 bearing nomenclature

The **fifth week** focused on practical maintenance tasks. I inspected a bridge-type reclaimer and assisted in repositioning a misaligned roller. I participated in clearing a blockage in a fuller pump nozzle, which deepened my understanding of fluid mechanics by observing how nozzle constrictions influence velocity and pressure. I witnessed filter bag replacement, which was necessary to control stack emissions, and observed bucket elevator inspections where

cracked stirrups were identified and replaced. I also participated in turbo coupling seal replacement followed by shaft alignment with technicians. These activities enhanced my appreciation for reliability practices and teamwork in maintenance.



Figure 7 cracked stirrup on the bucket

In the **sixth week**, I engaged in more specialized tasks. I observed inspections around vertical roller mills, participated in conveyor belt roller replacement, and assisted in sealing compressed air leaks on fuller pumps. I helped in replacing worn-out hammers on a hammer mill and attended a tutorial session with a reliability engineer on Mean Time Between Failures (MTBF). I was guided on aligning a crane drive system using a feeler gauge and introduced to laser alignment procedures. These sessions combined theoretical training with hands-on practice, improving both my technical and analytical skills.



Figure 8 fitting the shaft back that holds the hammers

The **seventh week** allowed me to participate in more advanced maintenance tasks. I assisted in cleaning a thermopac coil and combustion chamber, observed apron feeder inspections, and learned how filter fans are checked for pressure differentials. I participated in cooler fan maintenance, where I observed impeller cleaning, endoscopic inspection of bolts, and fan balancing using tachometers. I also gained a deeper understanding of rotary kiln operation, roller lubrication, and tire clearance measurements. Using a vernier caliper, I learned how looseness is assessed. Later, I observed thermographic analysis of rotary kiln components, which revealed how uneven lubrication on gears could be detected by irregular temperature readings.



Figure 9 rotary kiln roller lubrication

In the **eighth week**, I was exposed to both mechanical and utility systems. I observed conveyor roller replacements and packing plant maintenance, participated in clinker loading plant inspections, and assisted in dismantling Plummer blocks on elevators for bearing checks. I helped open screw conveyors for inspection of flange couplings and bearings. I also took part in preventive maintenance inspections of screws and bucket elevators. Towards the end of the attachment, I finalized clearance procedures, returned my PPE, and obtained my recommendation letter from the supervisor after bidding farewell to my team.

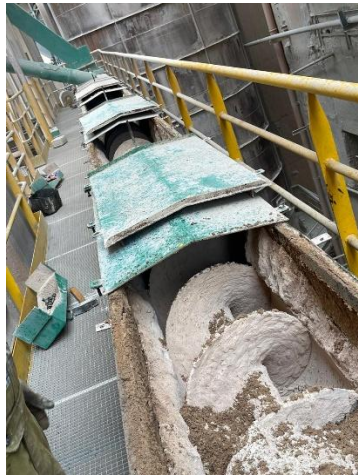


Figure 10 Screw conveyor inspection



Figure 11 Inside of two chamber ball mill

Individually, I contributed by tightening bolts, assisting in disassembly of compressors, gearboxes, and motors, and participating in inspection of mills, conveyors, and fans. I gained hands-on exposure to tools such as stroboscopes, feeler gauges, tachometers, and

thermocameras. As part of a team, I worked alongside technicians in repositioning rollers, cleaning fans, replacing hammers, and conducting alignment procedures. I also supported engineers and inspectors in troubleshooting equipment and participated in discussions on equipment reliability and maintenance planning.

CHAPTER 4 PERSONAL DEVELOPMENT AND AWARENESS ABOUT CONTEMPORARY ISSUES

4.1 Acquired Skills and Recognition of Contemporary Issues

My summer internship at Bamburi Cement in Mombasa was a profound period of professional and personal growth. It provided me with a practical, hands-on understanding of heavy industrial operations and allowed me to develop a wide range of technical and soft skills. Furthermore, it offered critical insight into the contemporary issues facing the modern manufacturing sector, particularly concerning environmental sustainability, operational efficiency, and technological integration.

1. Acquired Skills and Capabilities

During my attachment in the Mechanical Department, I gained a diverse set of skills:

Technical Maintenance and Inspection Proficiency: I actively participated in and observed numerous maintenance procedures. This included hands-on tasks such as disassembling high-volume screw compressors and high-torque gearboxes, assisting in the replacement of worn-out hammers on a hammer mill, and tightening bolts on conveyor systems and Plummer blocks. I became familiar with inspection routines for critical equipment like bucket elevators, rotary kiln rollers, ball mills, and separators, learning to identify signs of wear, misalignment, and potential failure.

Precision Alignment and Measurement Techniques: I gained valuable experience in precision engineering practices. I observed and assisted in shaft alignment procedures using both traditional tools like filler gauges and advanced technology like laser alignment systems. I also learned how to measure critical clearances, such as the 3mm gap between airlock vanes and housing, and kiln tire looseness using a vernier calliper.

Diagnostic and Analytical Skills: I was exposed to modern predictive maintenance techniques. I learned how vibration data is collected using triaxial accelerometers and analyzed through spectrograms to diagnose potential faults in bearings and other rotating components. I also observed the use of a thermocamera to monitor temperature profiles of

the kiln and girth gear, understanding that uneven temperature distribution can indicate issues like improper lubrication.

Understanding of Mechanical Systems: I developed a comprehensive understanding of the cement manufacturing process, from the crusher section and Pre-Blending Hall to the rotary kiln and packing plant. I learned about the function and maintenance of key systems, including the Induced Draft (ID) fan, Gas Conditioning Tower (GCT), vertical roller mills, and various conveyor and elevator systems.

Safety and Procedural Adherence: From the initial induction and "working at heights" training to daily shadowing of technicians, I ingrained a strong sense of safety culture. I learned the importance of Lockout-Tagout (LOTO) procedures, the correct use of Personal Protective Equipment (PPE), and the communication protocols essential for working safely in a high-risk environment.

Professional Soft Skills: The internship honed my ability to work effectively in a team, follow instructions from supervisors and senior technicians, and communicate technical observations clearly. The process of finalizing documentation and securing clearance also taught me valuable administrative and professional etiquette skills.

2. Recognition of Contemporary Issues

My experience at Bamburi Cement highlighted several key contemporary issues that are central to modern industrial operations:

Environmental Management and Emission Control: I gained direct awareness of the cement industry's environmental challenges and the technologies used to mitigate them. I witnessed the replacement of filter bags in the baghouse to reduce stack emissions, a critical action for controlling particulate matter. Furthermore, I learned about the role of the Gas Conditioning Tower (GCT) in performing evaporative cooling to reduce gas temperatures before treatment, which is essential for meeting environmental standards. My visit to the Reverse Osmosis (RO) plant underscored the importance of sustainable water management, ensuring that water used for evaporative cooling is free of impurities.

Operational Efficiency and Energy Consumption: The cement production process is highly energy-intensive. I observed a direct link between maintenance quality and energy efficiency. For instance, I learned how a blocked nozzle in a Fuller pump, or a misaligned conveyor roller,

leads to operational inefficiencies and increased energy consumption. The focus on predictive maintenance using vibration analysis and thermal imaging is a contemporary approach to prevent unplanned downtime, optimize Mean Time Between Failures (MTBF), and thereby reduce energy waste and operational costs.

Predictive Maintenance and Digitalization: The industry is rapidly moving from reactive to predictive maintenance models. My involvement in collecting and analysing vibration and temperature data demonstrated how data-driven insights are used to foresee equipment failures. The tutorial session on MTBF with a reliability engineer emphasized the importance of using data to predict equipment lifecycles and plan maintenance schedules, minimizing disruptive and costly breakdowns. This represents a significant shift towards Industry 4.0 principles within traditional manufacturing.

Resource Optimization and Circular Economy: While not explicitly stated in the logbook, working in a plant that uses raw materials like coral and manages by-products like cement kiln dust provides an inherent understanding of resource management. The focus on preventing material leaks (e.g., from packers or screw conveyors) and reusing materials, such as using a cut conveyor belt as a temporary seal, reflects a practical approach to waste minimization and resource optimization, which are pillars of the circular economy.

REFERENCES

- [1] Bamburi Cement Limited, "Bamburi Cement Sustainable Development Report 2022," Nairobi, 2023.
- [2] Bamburi Cement Limited, "Bamburi Cement Launches Eco-Building Solutions in Mombasa," Mombasa, 2024.
- [3] Holcim Group, "Holcim Integrated Report: Building Progress for People and the Planet," Zurich, 2023.
- [4] Bamburi Cement Limited, "Bamburi Cement Climate Report 2021: Our Pathway to Net Zero," Nairobi, 2022.
- [5] Mombasa County Government, "Partnership with Bamburi Cement for Water Conservation Project," Mombasa, 2023.

APPENDIX A STANDARDS

A. Quality Management Standards

ISO 9001:2015 - Quality Management Systems: This is the core standard ensuring that the company's processes consistently provide products that meet customer and regulatory requirements. It covers all aspects, from raw material sourcing and production (clinker making, cement milling) to packing and dispatch.

Kenya Bureau of Standards (KEBS) Standards: All cement products conform to specific Kenyan Standards (KS), such as:

KS EAS 18-1:2017 - Cement - Part 1: Composition, specifications and conformity criteria for common cements.

KS EAS 148-1:2017 - Cement - Part 1: Composition, specifications and conformity criteria for common cements (For CEM I Cement).

B. Environmental Management Standards

ISO 14001:2015 - Environmental Management Systems: This standard provides the framework for managing environmental responsibilities. Activities observed, such as stack emission monitoring (filter bag replacement), water treatment at the Reverse Osmosis (RO) plant, and the operation of the Gas Conditioning Tower (GCT) for emissions control, are all part of this systematic approach.

Environmental Management and Co-ordination Act (EMCA), 1999: The company complies with Kenyan national environmental regulations, including air quality, waste management, and water discharge standards set by the National Environment Management Authority (NEMA).

C. Occupational Health & Safety (OHS) Standards

ISO 45001:2018 - Occupational Health and Safety Management Systems: This is the international standard for occupational health and safety. The induction training, "Working at Heights" training, mandatory use of Personal Protective Equipment (PPE), and the daily safety

protocols followed during equipment inspections and maintenance are all integral components of this system.

Occupational Safety and Health Act (OSHA), 2007: Bamburi Cement adheres to the legal requirements for workplace safety as stipulated by Kenyan law.

D. Product-Specific and Performance Standards

ISO 19694-1:2021 - Stationary source emissions — Determination of greenhouse gas emissions in energy-intensive sectors — Part 1: This standard provides a general methodology for monitoring and reporting CO₂ emissions from cement production, which is critical for the company's environmental reporting.

Various ASTM International (ASTM) / European (EN) Standards: For specific product testing and performance characteristics, especially for specialized or export products, the company may reference additional international standards like those from ASTM or CEN.

E. Corporate Group Standards (Holcim)

Holcim Manufacturing Standards (HMS): As part of the global Holcim Group, Bamburi implements a set of stringent internal corporate standards that often exceed local regulatory requirements. These cover every aspect of operations, including:

Safety Rules: Defining minimum safety requirements for tasks like Lockout-Tagout-Try-out (LOTOTO), confined space entry, and working at height.

Technical Standards: Specifying maintenance procedures, predictive maintenance protocols (like vibration analysis and thermography), and equipment reliability metrics (like Mean Time Between Failures - MTBF).

Environmental & Sustainability Performance Indicators: Setting group-wide targets for CO₂ reduction, alternative fuel usage, and water conservation.

APPENDIX B CURRICULUM VITAE

TARIK TARIQ

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OBJECTIVE

Searching for a demanding job with a forward-thinking company that will allow me to use my technical expertise, contribute to organizational operations, and support my professional and personal development.

EDUCATION

2016 – 2020 **EASTERN MEDITERRANEAN UNIVERSITY, TURKEY**

BS Mechanical Engineering, CGPA 3.61, (ABET Accredited Program)

2012 – 2016 **NAISULA SCHOOL, KENYA**

PROJECTS

2025 **Indirect Evaporative Cooler**

2024 **Tad Pole Tricycle Based on ASME Guidelines**

2023 **Peaucellier-Lipkin Mechanism**

2017 **Course Project**

EXPERIENCE

Jul 2016 – Sep 2016 **Bamburi Cement**

Position: *Trainee Engineer (internship)*

Job Description: Inspection, Maintenance, Material Handling, and Planning.

ACHIEVEMENTS

- Received a Honors certificate for 2023-2024 fall semester.
- Received a Higher Honors certificate for 2023-2024 spring semester.
- Received a Higher Honors certificate for 2024-2025 fall semester.
- Received a Higher Honors certificate for 2024-2025 spring semester.

SKILLS

Software: ANSYS, CAD Modelling and Assembly in SolidWorks, and MATLAB.

Management: Work Breakdown Structure (WBS), Quality Function Deployment (QFD), Failure Mode & Effect Analysis (FMEA), Total Quality Management (TQM), Lean & Six Sigma

Microsoft Office: MS Word, MS PowerPoint, MS Excel, MS Project

Language: Proficient in English and Swahili